CS24204 IT WORKSHOP

Smart City Environmental Monitoring and Sustainability Analytics System

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Introduction

Smart cities in India are unable to monitor their environment and sustainability as a whole, leading to several environmental issues and low quality of life.

Traffic departments record vehicle flow, pollution control boards record air quality, meteorological agencies record weather, and utilities record energy use. Because each dataset has its own format, sampling rate and owner, planners rarely see how these factors interact. For example, a PM2.5 spike may be blamed on traffic when low wind speed or high energy demand is also at play.

Without an integrated view, cities cannot easily:

- identify which factors drive pollution at different times of day and seasons
- measure sustainability with a single comparable indicator
- forecast pollution early enough to issue warnings or adjust traffic and energy policy

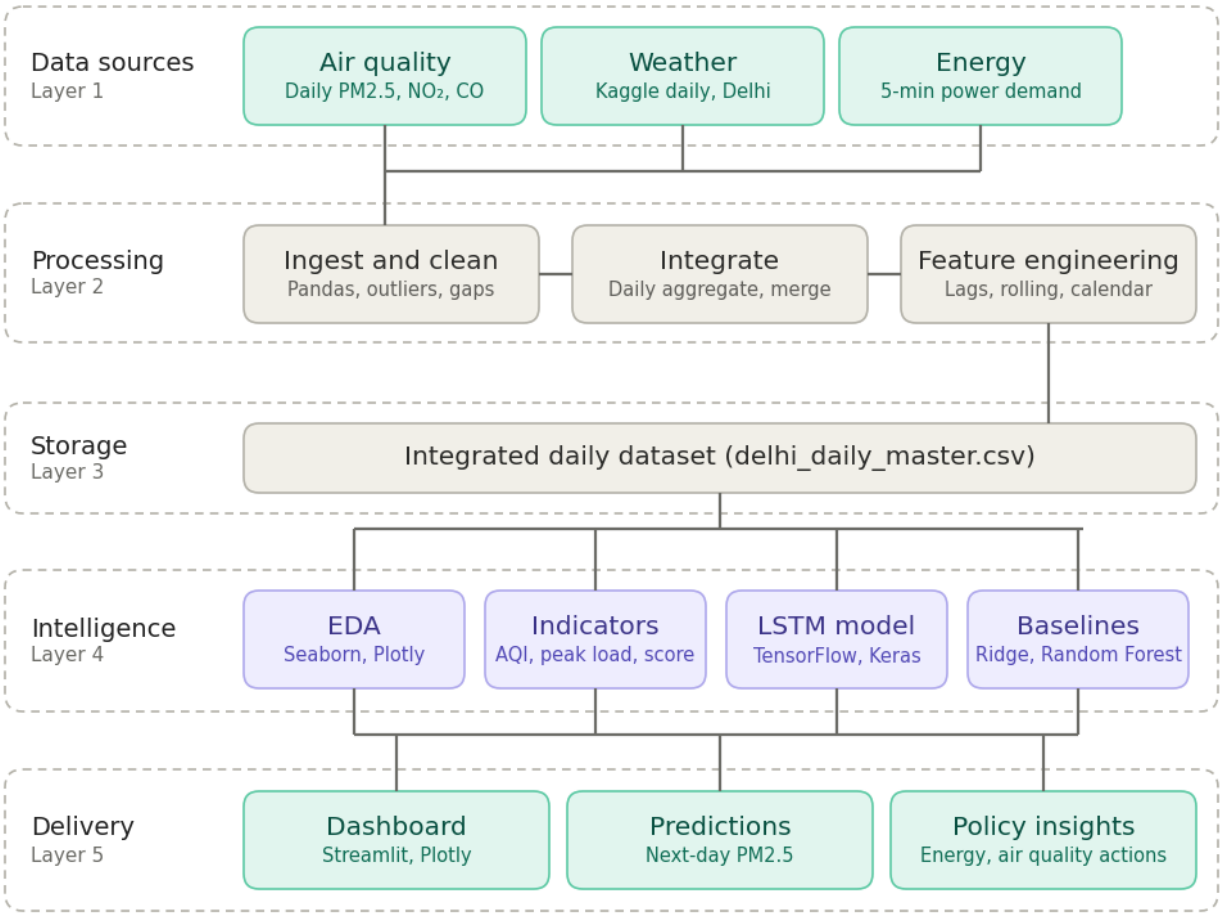
Objectives

1. To collect and integrate Delhi air quality, weather and energy demand datasets onto a common daily time scale.
2. To clean the data and engineer features such as weekday, weekend and holiday flags, season, lag values and rolling averages.
3. To analyze relationships among weather, energy demand and pollutants, and to compute sustainability indicators (AQI category distribution, peak-load ratio and a 0-100 Sustainability Score).
4. To develop an LSTM-based TensorFlow model that predicts next-day PM2.5 and compare it with persistence, Linear Regression and Random Forest baselines.
5. To derive practical insights that city planners can use for energy planning and air quality alerts.

Scope and Limitations

The study covers Delhi at daily resolution from 1 January 2023 to 30 November 2024 (700 days) and integrates air quality, weather and energy demand. Traffic data was left out because no suitable daily traffic series for Delhi was available. The air quality data is probably modelled rather than measured by ground sensors, and 700 days is a small sample for a deep learning model, so simple baselines are expected to perform comparably.

System Design



Data Processing

Summary

Item	Value
City	Delhi, India
Resolution	Daily
Study window	1 Jan 2023 to 30 Nov 2024 (700 days)
Domains	Air quality, weather, energy demand
Prediction target	Next-day PM2.5 (µg/m^3)

Source datasets used

Air quality (daily)

Property	Detail
Period	1 Jan 2023 to 31 Dec 2025
Rows and columns	1,096 rows, 18 columns
Variables	PM2.5, PM10, CO, NO_2, SO_2, O_3, dust, US AQI, European AQI, city, coordinates
Missing values	None

PM2.5 shows the correct Delhi seasonality, averaging about 121 µg/m^3 in January and 109 in December against 44 to 49 in July to September. PM2.5 falls with wind speed and temperature, which is physically correct. Day-to-day autocorrelation is 0.85, and PM2.5 correlates with NO_2 (0.61) and CO (0.76).

Weather (daily) for Delhi

Property	Detail
Period	14 Apr 2013 to 30 Nov 2024
Rows and columns	3,557 rows, 29 columns
Variables used	Temperature, humidity, precipitation, wind speed, sea-level pressure

Energy demand (5-minute)

Property	Detail
Period	1 Jan 2021 to 12 Dec 2024
Rows and columns	393,440 rows, 15 columns
Variables	Power demand (probably MW), temperature, dew point, humidity, wind direction and speed, pressure, calendar fields

Processing pipeline

1. Load and clean each source: remove index columns, drop duplicates, drop derived and leakage-prone columns, treat wind outliers.
2. Aggregate energy demand from five-minute readings to daily mean and daily peak, applying the 80% coverage rule and interpolating the days that fail it.

3. Merge the three sources on date and restrict to the overlap window (1 Jan 2023 to 30 Nov 2024).
4. Engineer features: calendar fields, holiday flag, October to November season flag, lags of 1, 2 and 7 days, 7-day rolling PM2.5 mean and 3-day rolling temperature mean. Lags use only past values.
5. Compute indicators: AQI category and the sustainability score.
6. Split chronologically, fit the scaler on training days only, and build 7-day windows.

Final dataset: data dictionary

The integrated table delhi_daily_master.csv has one row per day and 34 columns.

Group	Columns	Meaning
Date	date	Calendar day
Air quality	pm2_5, pm10, co, no2, so2, o3, dust_log	Daily pollutant concentrations; dust is log-transformed
Weather	temp, humidity, precip, windspeed, pressure	Daily weather
Energy	demand_mean, demand_peak, energy_coverage, peak_load_ratio, energy_interpolated	Daily average and maximum demand, share of readings present, peak divided by mean, flag for interpolated days
Calendar	weekday, month, is_weekend, season, stubble_season, month_sin, month_cos, is_holiday	Time and season indicators; holidays come from the Python holidays package
Engineered	pm2_5_lag1, pm2_5_lag2, pm2_5_lag7, pm2_5_roll7, temp_roll3, target_next_day_pm25	Past-value features and the prediction target
Indicators	aqi_category, sustainability_score	Sustainability indicators (see Indicators below)

The model uses 18 input features: PM2.5, PM10, NO_2, CO, SO_2, O_3, log dust, temperature, humidity, wind speed, precipitation, pressure, mean demand, peak demand, weekend flag, holiday flag, and the month sine and cosine.

Summary of the integrated data

Variable	Mean	Min	Max
PM2.5 (µg/m^3)	64.5	15.3	214.0
Temperature (°C)	25.6	8.3	39.5
Wind speed (km/h)	15.0	5.4	59.4
Mean demand (MW)	4,187.9	2,121.3	7,361.1
Peak demand (MW)	5,154.9	2,493.1	8,631.5
Peak-load ratio	about 1.2	about 1.0	about 1.5

Correlation with PM2.5 over the study window: temperature -0.60, wind speed -0.44, mean demand

-0.28, humidity +0.27, precipitation -0.13.
AQI category distribution (daily PM2.5, CPCB breakpoints): Satisfactory 366 days, Moderate 151, Poor 84, Very Poor 59, Good 40, Severe 0.

Indicators

- AQI category: Good up to 30, Satisfactory 31 to 60, Moderate 61 to 90, Poor 91 to 120, Very Poor 121 to 250, Severe above 250 (µg/m³, 24-hour PM2.5).
- Peak-load ratio: daily peak demand divided by daily mean demand.
- Sustainability score (0 to 100, higher is better): 100 minus 100 times the weighted sum of normalised PM2.5 (weight 0.50), normalised mean demand (0.30) and normalised weather stress (0.20). Weather stress is the absolute distance of temperature from 24 °C. These weights are a design choice and must be justified in the report. Over the study window the score has a mean of 67.8 and a range of 12.8 to 90.9.

Modelling setup

Item	Setting
Target	PM2.5 on the day after the window
Input	7 days by 18 features
Split	Chronological 70/15/15
Windows	482 train, 105 validation, 106 test
Split dates	Train ends 3 May 2024; validation ends 16 Aug 2024; test runs to 30 Nov 2024
Scaling	Min-max scaler fitted on training days only
Baseline to beat	Persistence (tomorrow equals today): test RMSE 17.9 µg/m³

Results and Discussion

Model comparison

Table 1 compares the models on the test period (16 Aug to 30 Nov 2024, 106 days). RMSE and MAE are in µg/m³.

Model	RMSE	MAE	R^2
Persistence (baseline)	18.0	13.0	0.75
Linear Regression (Ridge)	17.5	12.6	0.77
Random Forest	17.2	12.5	0.78
LSTM	17.0	12.3	0.78

Table 1: Test-set performance of all models.

Predicted versus actual

Figure 1 plots the LSTM predictions against the observed PM2.5 over the test period, and Figure 2 shows the model comparison.

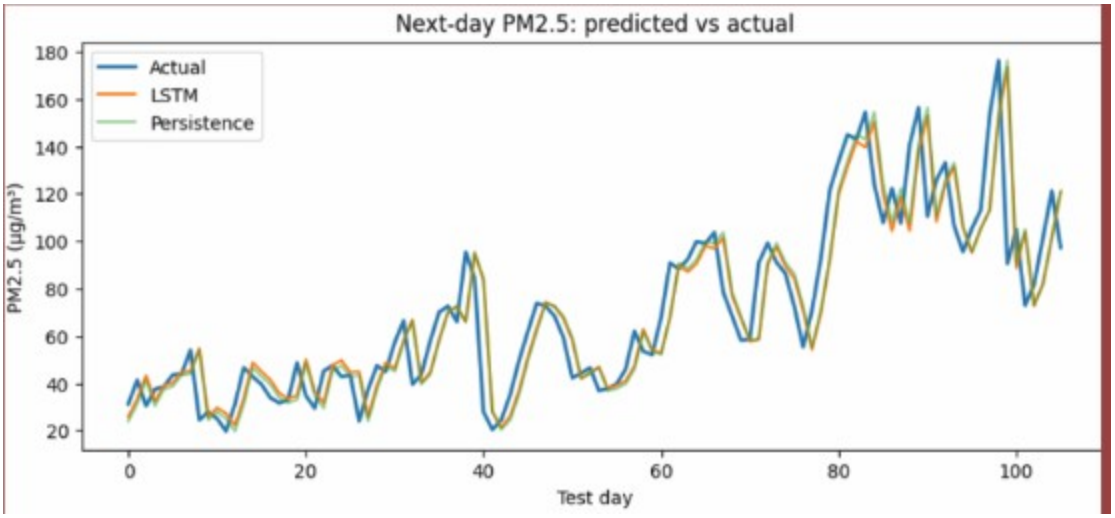


Figure 1: LSTM Prediction Plot

	RMSE	MAE	R2
All features	17.659	12.749	0.768
Without weather	17.634	12.743	0.768
Without energy	17.549	12.679	0.771
Without other pollutants	17.388	12.755	0.775
PM2.5 history only	16.873	12.422	0.788

Figure 2: model comparison

Effect of feature groups (ablation)

To test whether weather, energy and other pollutants help the forecast, the LSTM was retrained with each group removed (Table 2).

Feature set	RMSE	MAE	R^2
All features	17.659	12.749	0.768
Without weather	17.634	12.743	0.768
Without energy	17.549	12.679	0.771
Without other pollutants	17.388	12.755	0.775
PM2.5 history only	16.873	12.422	0.788

Table 2: Ablation of feature groups (LSTM).

Conclusion

On the test period (16 August to 30 November 2024), all models performed within about 4% of each other. Persistence reached an RMSE of 17.94 µg/m³, Ridge regression 17.43, Random Forest 17.20 and the LSTM 17.66 (R^2 of 0.76 to 0.78). The LSTM beat persistence but not the Random Forest, which is expected given only about 480 training days. The feature ablation showed that PM2.5 history carries most of the predictive information, and that weather and energy inputs add little to a next-day forecast on this much data. Their main value lies in the sustainability indicators and the seasonal insight that Delhi's electricity demand peaks in summer while its pollution peaks in winter.

Limitations

- The air quality data is probably modelled rather than measured, and covers a single location.
- Only 700 days overlap across all sources, which is small for a deep learning model.
- The test period covers the autumn build-up to winter pollution, which training only partly represents.
- Traffic is excluded, so the system integrates three domains rather than four.
- Nineteen energy days in the study window were interpolated, and weather is available only at daily resolution.

References

[1] Y. Li et al., "Urban PM2.5 concentration prediction via attention-based CNN-LSTM," Applied Sciences, 2020.
[2] Thaweephol and Wiwatwattana, "Long short-term memory deep neural network model for PM2.5 forecasting in the Bangkok urban area," 2019.
[3] Surmeli and Alptekin, "Modeling the relationship between traffic intensity and urban air pollution with LSTM networks," IntelliSys, 2025.
[4] Tiwari, "The machine learning framework for traffic management in smart cities," Management of Environmental Quality, 2023.
[5] "Deep learning based multimodal urban air quality prediction and traffic analytics," Scientific Reports, 2023.

[6] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997.
[7] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.
[8] M. Abadi et al., "TensorFlow: Large-scale machine learning on heterogeneous systems," arXiv:1603.04467, 2016.
[9] Central Pollution Control Board (India), "National Air Quality Index," CPCB.